EEE301 Signals and Systems
Tutorial 5: Analysis of LTI Systems in the s-Domain

1. By direct integration, find the Laplace transforms for:
   i) $te^{-t}u(t)$ ii) $(e^{2t}-2e^{-t})u(t)$

2. Find the inverse unilateral Laplace transforms of
   i) $\frac{2s+5}{s^2+5s+6}$ ii) $\frac{(s-1)^2}{s^2-s-6}$
   iii) $\frac{2s+1}{(s+1)(s^2+2s+2)}$ iv) $\frac{s+2}{s(s+1)^2}$

3. Find the Laplace transforms of the following using a Laplace transform table and Laplace transform properties:
   i) $e^{-(t-\tau)}u(t)$ ii) $te^{-t}u(t-\tau)$ iii) $\sin[\omega_0(t-\tau)]u(t)$

4. Using Laplace transform, solve the following differential equations and determine the zero input and zero state components.
   i) $\frac{d^2y}{dt}+3\frac{dy}{dt}+2\,y(t)=\frac{df}{dt}$ if $y(0^-)=y'(0^-)=0$ and $f(t)=u(t)$
   ii) $\frac{d^2y}{dt}+4\frac{dy}{dt}+4y(t)=\frac{df}{dt}+f(t)$ if $y(0^-)=2$, $y'(0^-)=1$ and $f(t)=e^{-t}u(t)$

5. For the system with transfer function $H(s)=\frac{s+5}{s^2+5s+6}$
   i) Write the differential equation relating output $y(t)$ to input $f(t)$
   ii) Find the zero state response if the input $f(t)=e^{-4t}u(t)$